

Tutorial-4

1. Given: pore radius, $r = 50 \text{ \AA}$
 $T = 100^\circ\text{C}$
 $p = 1 \text{ or } 10 \text{ atm}$
 $\epsilon = 0.4$
 $\sigma = 0.8$
 $\tau = 4$

a) $r = 50 \times 10^{-8} = 5 \times 10^{-7} \text{ cm}$

$T = 373.15 \text{ K}$

$D_{AB} = 0.86/p \text{ cm}^2 \text{ sec}^{-1}$

$D_{K_{H_2}} = 9.7 \times 10^3 (5 \times 10^{-7}) \sqrt{373.15/2.016}$
 $= 0.0656 \text{ cm}^2/\text{sec}$

The molecular mass is 2.016 gm/mol . ~~Kn~~ Knudsen diffusivity is independent of pressure in the supplied correlation, because transport is controlled by collisions with the pore walls.

b) $\frac{1}{D_{H_2}} = \frac{1}{D_{AB}} + \frac{1}{D_K}$

$D_{H_2} = \frac{D_{AB} D_K}{(D_{AB} + D_K)}$

c) $D_e = D_{H_2} \epsilon \sigma / \tau = D_{H_2} (0.4 \times 0.8 / 4)$
 $= 0.08 D_{H_2}$

at 1 atm $D_{AB} = 0.86$ $D_K = 0.066$ $D_{H_2} = 0.061$

$D_e = 0.005$

at 10 atm $D_{AB} = 0.086$ $D_K = 0.066$ $D_{H_2} = 0.037$

$D_e = 0.003$

Teacher's Signature.....

Q.2 2nd order, $d_p = 0.5 \text{ mm}$, $\rho_p = 2.2 \text{ g/cm}^3$, $q_m = 10^5 \text{ mol/h kg-cat}$
 $C_A = 20 \text{ mol/m}^3$, $D_e = 5 \times 10^{-5} \text{ m}^2/\text{h}$

Using the radius-based Weisz-Prater parameter (C_{WP})

$$C_{WP} = \frac{q_m \rho_p R^2}{q D_e C_{AB}}$$

$$C_{WP} \approx \frac{(2.2 \times 10^8) (2.5 \times 10^{-4})^2}{9 \times 5 \times 10^{-5} \times 20} = \frac{13,750}{9} = 1528$$

$C_{WP} \gg 1$. If the given concentration is the bulk value rather than the ϕ surface value, $C_{AB} \leq 20$ and the actual parameter is even larger.

Strong pore diffusion resistance is indicated.

Q.3 (Similar to Fogler Eg. 15-2)

1st order; $\rho_p = 1.6 \text{ gm/cm}^3$, $D_e = 2 \times 10^{-6} \text{ m}^2/\text{sec}$

Run 1: $q_{m1} = 3 \times 10^{-5} \text{ mol/g-cat sec}$, $R_1 = 0.01 \text{ m}$

Run 2: $q_{m2} = 15 \times 10^{-5} \text{ mol/g-cat sec}$; $R_2 = 0.001 \text{ m}$

$q_m = \eta k C_{AB} / \rho_p$. Taking the ratio cancels both k and C_{AB}

$$\eta(\phi) = 3(\phi \coth \phi - 1) / \phi^2$$

$$\phi_1 = \frac{R_1}{R_2} \phi_2 = 10 \phi_2$$

$$\frac{\eta(10\phi_2)}{\eta(\phi_2)} = \frac{q_{m1}}{q_{m2}} = \frac{3}{15} = 0.2$$

Teacher's Signature.....

kg-cat

$$R = 10 \text{ mm} \quad \phi = R \sqrt{K/D_e} = 16.46$$

$$\eta = 0.17 \quad C_{wp} = \eta \phi^2 = 46.3684$$

$\therefore$ Strong pore diffusion

$$R = 1 \text{ mm} \quad \phi = 1.65 \quad \eta = 0.86$$

$$C_{wp} = \eta \phi^2 = 2.32$$

$\therefore$ Moderate internal effect ; not strong

Q.4 $C_{As} = 0.001 \text{ g mol/dm}^3$ $d = 0.2 \text{ cm}$ $R = 0.1 \text{ cm}$
 $D_e = 0.1 \text{ cm}^2/\text{sec}$

$$\eta = R - 0.03 = 0.07 \text{ cm and } \xi = R/\eta = 0.7$$

$$\phi = R \sqrt{K/D_e}$$

a) $C_A/C_{As} = \sinh(\xi \phi) / \xi \sinh \phi$

$$\phi = 0.1 \text{ cm} \sqrt{K/0.1 \text{ cm}^2 \text{ sec}^{-1}}$$

$$C_A = \frac{0.001}{0.7} \sinh(0.7\phi) / \sinh \phi \quad \text{mol dm}^{-3}$$

$$(\eta = 0.07 \text{ cm})$$

$$= 9.99 \times 10^{-4} \text{ mol/dm}^3$$

b) $0.8 = 3(\phi_{\text{new}} \coth \phi_{\text{new}} - 1) / \phi_{\text{new}}^2$
 $\phi_{\text{new}} = 2.042 \quad \phi = \frac{\phi_{\text{new}}}{3} = 0.68$

$$d_{\text{new}} = 2\phi_{\text{new}} \sqrt{D_e/K} = 4.084$$

Teacher's Signature.....

Q.5 $d_p = 2.4 \text{ mm}$ $D_e = 5 \times 10^{-5} \text{ m}^2 \text{ h}^{-1}$ $K_g = 300 \text{ mh}^{-1}$
 $C_{Ag} = 20 \text{ mol m}^{-3}$ $q_v = 10^5 \text{ mol h}^{-1} \text{ m}^{-3} \text{-cat}$
 first order, Density = 2.8 g cm^{-3}
 pellet porosity = 0.4
 $R = 1.2 \times 10^{-3} \text{ m}$
 $V_p / S_p = R/3 = 4 \times 10^{-4} \text{ m}$

$$C_M = \frac{n q_v R}{K_g C_{Ag}} = \frac{1 \times 10^5 \times 1.2 \times 10^{-3}}{300 \times 20}$$

$$= 0.02 < 0.15$$

Meares Criterion for external mass transfer

$\therefore$ External mass transfer resistance is negligible.

$$q_v V_p = K_g S_p (C_{Ag} - C_{As})$$

$$C_{As} = 20 - q_v R / 3 K_g = 19.87 \text{ mol m}^{-3}$$

$$(C_{Ag} - C_{As}) / C_{Ag} = 0.667\%$$

$$C_{WP} = q_v R^2 / (D_e C_{As})$$

$$= \frac{10^5 (1.2 \times 10^{-3})^2}{(5 \times 10^{-5}) 19.87} \approx 14.9145 > 1$$

Weisz-Prater Criterion for internal diffusion

The observed rate is limited by internal pore diffusion; external mass transfer is negligible.

Teacher's Signature.....

Q.6 Assume the same catalyst mass, inlet flow, temperature and intrinsic kinetics, with unchange relevant concentration - conversion relation and negligible pressure effects on the rate. Under strong internal diffusion, $\eta \propto 1/r$ $\eta \propto 1/d_p$, so doubling the diameter halves the observed first-order rate coefficient.

$$K_{app,2} / K_{app,1} = d_{p1} / d_{p2} = 9/18 = 0.5$$

For a constant-density plug-flow model: $-\ln(1-X) = K_{app} \tau_0$

$$-\ln(1-X_2) = \frac{1}{2} [\ln(1-0.632)]$$

$$1-X_2 = \sqrt{0.368} = 0.61$$

$$X_2 = 0.39$$

Conversion falls from 63.2% to ~~about~~ about 39%. This a drop of 24.2% points. Halving the rate coefficient does not halve the conversion because the plug-flow relation is exponential.

Teacher's Signature.....